

Teaching Exam

Subject – Computer Science

Topic – Practice and Revision

Lecture No.- 02



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Topics to be Covered

Topic

1:- Practice & Revision Set



1. When the sender and the receiver of an email are on different systems, we need only _____

- a) One MTA
- b) Two UAs
- c) Two UAs and one MTA
- d) Two UAs and two MTAs

2. SMTP uses which of the following TCP port?

- a) 22
- b) 23
- c) 21
- d) 25

1. Answer: a

Explanation: Since TCP is a reliable protocol, it's more efficient to use TCP protocol for e-mail transfer. TCP also provides more security than other transport layer protocols.

2. Answer: d

Explanation: Port 21 is used for FTP control connection, port 22 is used by SSH, and port 23 is used by TELNET.

3. Which of the following protocol is not used to receive mail messages?

- a) SMTP ✓
- b) Post Office Protocol (POP) ✓
- c) Internet Message Access Protocol (IMAP) ✓
- d) FTP → File Transfer Protocol

4. What is on-demand mail relay (ODMR)?

- a) protocol for SMTP security
- b) ✓ an SMTP extension
- c) protocol for web pages
- d) protocol for faster mail transfer

Answer 3: d

Explanation: FTP is used to share files. SMTP, POP and IMAP are the protocols used to send and receive mails on the internet.

Answer 4: b

Explanation: ODMR is an extension to SMTP, in which mails are relayed to the receivers after they are authenticated. It allows only the authorized receivers to receive the mail.

5. An email client needs to know the _____ of its initial SMTP server.

- a) IP address ✓
- b) MAC address
- c) URL
- d) Name

6. SMTP defines _____

- a) message transport
- b) message encryption ✓
- c) message content
- d) message password

Message Transport
Security protocol for encryption
SMTP + SSL/TLS

Answer 5: a

Explanation: The client needs to know the IP of its initial SMTP server as it has to send the mail first to that server and then the server forwards the mail ahead on behalf of the user.

Answer 6: a

Explanation: As the name suggests, Simple Mail Transfer Protocol is only responsible for “how” the message is transferred i.e. Transport of the message. Other protocols such as TCP are used to provide other services like encryption for the messages.

7. The entire hostname has a maximum of _____

- a) 255 characters
- b) 127 characters
- c) 63 characters
- d) 31 characters

8. A DNS client is called _____

- a) DNS updater
- b) DNS resolver ✓
- c) DNS handler
- d) none of the mentioned

Answer 7: a

Explanation: An entire hostname can have a maximum of 255 characters. Although each label must be from 1 to 63 characters long. Host name is actually a label that is given to a device in a network.

Answer 8: b

Explanation: DNS client also known as DNS resolver also known as DNS lookup helps to resolve DNS requests using an external DNS server.

9. Servers handle requests for other domains _____

- a) directly
- ☒ b) by contacting remote DNS server
- c) it is not possible
- d) none of the mentioned

10. DNS database contains _____

- ☒ a) name server records
- ☒ b) hostname-to-address records
- ☒ c) hostname aliases
- ☒ d) all of the mentioned

Answer 9: b

Explanation: Whenever a request is received at server from other domains, it handles this situation by contacting remote DNS server.

Answer 10: d

Explanation: Domain Name system not only deals with mapping IP addresses with the hostname but also deals with exchange of information in the server.

11. If a server has no clue about where to find the address for a hostname then _____

- a) server asks to the root server
- ☒ b) server asks to its adjacent server
- c) request is not processed
- d) none of the mentioned

12. Which one of the following allows client to update their DNS entry as their IP address change?

- ☒ a) dynamic DNS
- b) mail transfer agent
- c) authoritative name server
- d) none of the mentioned

Answer 11: a

Explanation: Root name servers are actually very important and critical as they are the first step in translating human readable hostnames into IP addresses for carrying out communication.

Answer 12: a

Explanation: Dynamic DNS or in short DDNS or DynDNS helps in automatically updating a name server in the DNS. This does not require manual editing.

13. Wildcard domain names start with label _____

- a) @
- ☒ b) *
- c) &
- d) #

14. The domain name system is maintained by _____

- ☒ a) distributed database system
- b) a single server
- c) a single computer
- d) none of the mentioned

Answer 13: b

Explanation: A wildcard DNS record matches requests to a non existent domain name. This wildcard DNS record is specified by using asterisk "*" as the starting of a domain name.

Answer 14: a

Explanation: A domain name system is maintained by a distributed database system. It is a collection of multiple, logically interrelated databases distributed over a computer network.

15. Which one of the following is not true?

- a) multiple hostnames may correspond to a single IP address
- b) a single hostname may correspond to many IP addresses
- c) a single hostname always corresponds to a single IP address
- d) none of the mentioned

16. Which component is included in IP security?

- ~~a) Authentication Header (AH)~~ ✓
- ~~b) Encapsulating Security Payload (ESP)~~ ✓
- ~~c) Internet key Exchange (IKE)~~
- d) All of the mentioned

Answer15: c

Explanation: A single hostname does not always correspond to a single IP address. In many cases, a hostname can be associated with multiple IP addresses, such as in load balancing or Content Delivery Networks (CDNs). However, multiple hostnames can map to a single IP, and a hostname can have multiple IPs. Answer 14: a

Explanation: A domain name system is maintained by a distributed database system. It is a collection of multiple, logically interrelated databases distributed over a computer network.

Answer16: d

Explanation: AH ensures that there is no retransmission of data from an unauthorized source, and protects against data tampering. ESP provides with content protection and ensures that there is integrity and confidentiality for the message. IKE is used to make sure that only the intended sender and receiver can access the message.

17. WPA2 is used for security in _____

- a) Ethernet
- b) Bluetooth
- c) Wi-Fi ✓
- d) Email

18. Pretty good privacy (PGP) is used in _____

- a) Browser security
- b) Email security ✓
- c) FTP security
- d) WiFi security

Answer17: c

Explanation: WPA2 or WiFi Protected Access 2 is a security protocol used to provide users and firms with strong data security and protection for their wireless networks (WiFi) to give them confidence that only authorized users can access their network.

Answer18: b

Explanation: PGP is an encryption method used in e-mail security to encrypt and decrypt the content of an e-mail transmitted over the internet. It makes sure that the message cannot be stolen by other unauthorized users.

19. Extensible authentication protocol is authentication framework frequently used in _____

EAP

- a) Wired personal area network
- b) Wireless networks ✓
- c) Wired local area network
- d) Wired metropolitan area network

20. Each NIC has a _____ address, which helps in uniquely identifying the computer.

- 1. IP Address
- 2. Website Address
- 3. Email Address
- 4. MAC Address ✓

* UDP Protocol does not give guarantee that Data can reach safely from Sender to Receiver (User Datagram Protocol)

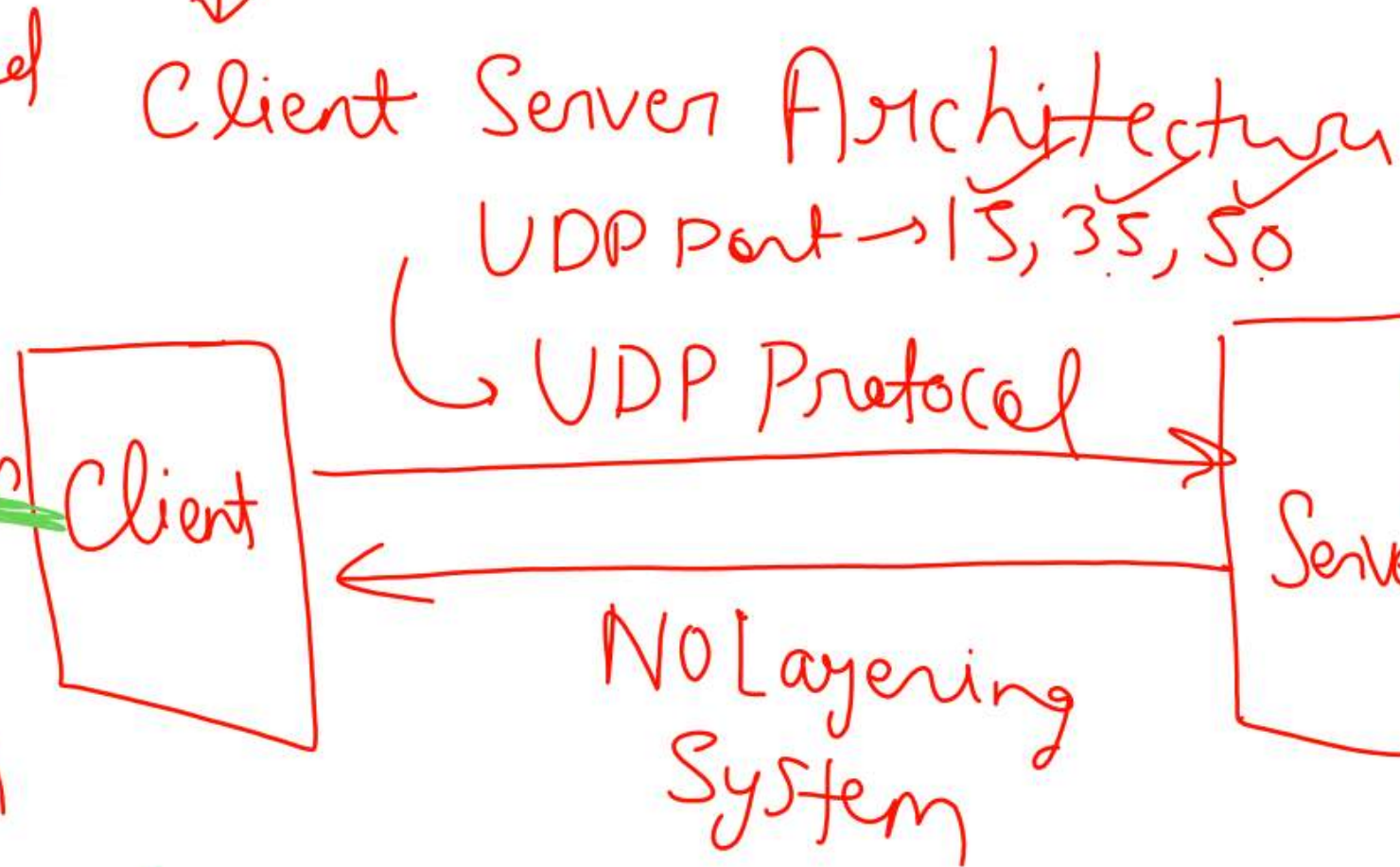
UDP → Transmission Protocol

Based on Client Server Architecture

TCP/IP
↳ Data Packet
20-60 byte

* for Transmitting fast Message Client to Server or Sender to Receiver

* Data Error Check → Checksum
* No Encryption Protocol



UDP Protocol
↳ Datagram → 8 byte

Uses

* Small message or Notification push
* Video Calling

THANK YOU